

VISIONINSPECT AI

AI-Powered Manufacturing Defect Detection & Quality Inspection System

A real-time deep computer vision and deterministic quality decision intelligence platform engineered for automated surface defect localization, multi-factor severity assessment, and scrap reduction.



Project Domain:	Automated Optical Inspection (AOI) / Industry 4.0
Deep Learning Engine:	Ultralytics YOLOv8 Defect Detection & Crop Classifier
Architecture:	FastAPI (Python) REST Backend & Next.js 16 Web Dashboard
Benchmark Dataset:	MVTec Anomaly Detection (15 Categories, 73 Defect Classes)
Documentation Type:	Final Major Project Technical Report & System Specification

1. PROJECT OVERVIEW AND OBJECTIVES

1.1 Industrial Context & Purpose

VisionInspect AI is an enterprise-grade automated optical inspection (AOI) platform engineered for modern manufacturing lines. High-volume manufacturing requires continuous visual quality verification to intercept surface flaws before packaging. VisionInspect AI eliminates manual inspection fatigue by uniting deep learning defect localization (YOLOv8) with a deterministic 4-state quality decision engine.



Figure 1.1: VisionInspect AI End-to-End Operational Workflow

1.2 Major Objectives & Expected Benefits

- **Automated High-Speed Inspection:** Delivers real-time inference matching conveyor throughput (<100ms).
- **Accurate Defect Localization:** Identifies flaw bounding boxes and types across 73 industrial classes.
- **Objective Multi-Factor Severity:** Calculates continuous risk scores (0–100) from size, location, and hazard.
- **Deterministic Quality Routing:** Enforces strict PASS/FAIL/REVIEW/REWORK rules to eliminate scrap waste.

2. PROBLEM STATEMENT AND SCOPE

2.1 Manufacturing Challenges & Proposed Solution

Manual visual inspection suffers from cognitive fatigue, subjective inconsistency, and high defect escape rates. Conventional optical systems rely on rigid binary thresholds that scrap repairable parts. VisionInspect AI resolves these challenges through decoupled deep learning defect detection and multi-factor quality routing.

IN-SCOPE CAPABILITIES (Implemented) • 15 MVTec AD Categories & 73 Defect Classes • Multi-Factor Severity Scoring (0.0 to 100.0) • Deterministic 4-State Routing (PASS/FAIL/REV/REW) • Role-Based Dashboards & Audit Trail Logging	OUT-OF-SCOPE / FUTURE BOUNDARIES • Direct Hardware PLC / SCADA Modbus bus • Automated physical pneumatic robotic arms • 3D volumetric laser or thermal/X-ray scans • Online edge continuous self-retraining
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Figure 2.1: Defined Project Scope and Operational Boundaries

2.2 Project Scope Breakdown

- **Implemented Scope:** Surface defect localization across 15 MVTec AD categories, 73 defect classes, multi-factor severity scoring, deterministic 4-state routing, role-based dashboards, and containerized Docker deployment.
- **Out-of-Scope / Future Work:** Direct PLC/SCADA Modbus hardware bus integration, automated physical pneumatic ejection arms, 3D volumetric laser scanning, and online edge self-retraining.

3. SYSTEM ARCHITECTURE

3.1 Decoupled Multi-Tier Architecture

VisionInspect AI is structured as a decoupled multi-tier architecture separating deep learning inference from deterministic manufacturing quality evaluation, relational persistence, and role-based presentation services.

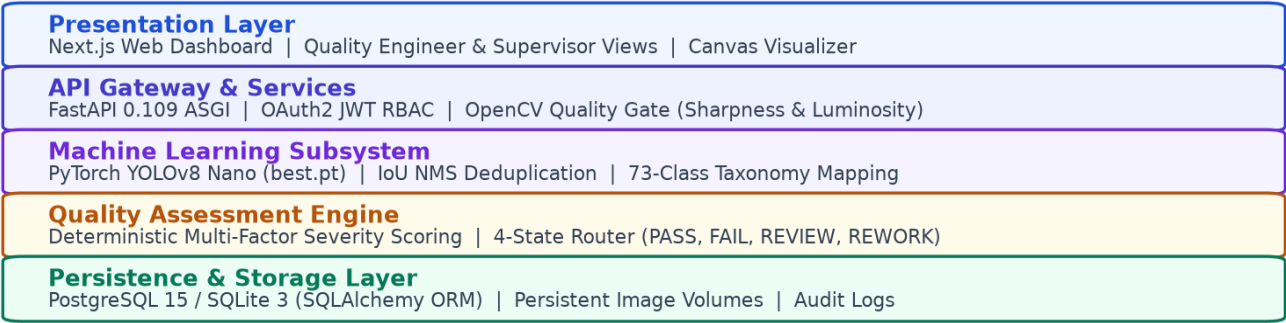


Figure 3.1: VisionInspect AI Decoupled System Architecture & Layer Topology

3.2 Architectural Reliability & Gateways

- **Edge Quality Gate:** OpenCV evaluates Laplacian sharpness variance and luminosity to filter blurry images.
- **Zero-CORS Reverse Proxy:** Nginx routes `/`` to Next.js and `/api/*`` to FastAPI, eliminating CORS preflight latency.
- **Transactional Persistence:** SQLAlchemy ORM manages structured persistence across PostgreSQL and SQLite.

4. TECHNOLOGY STACK

4.1 Verified Technology Stack Matrix

The system utilizes enterprise-grade open-source frameworks across presentation, API, machine learning, and database layers:

Frontend	Backend API	AI / Vision	Database	Deployment
Next.js 16 React 19 TypeScript Tailwind CSS	FastAPI 0.109 Python 3.11 SQLAlchemy Uvicorn ASGI	PyTorch 2.0+ Ultralytics YOLOv8 OpenCV (`cv2`) NumPy	PostgreSQL 15 SQLite 3 Relational Schema Audit Trail	Docker Compose Multi-stage Builds Nginx Proxy AWS Topology

Figure 4.1: Categorized Technology Stack Architecture

4.2 Technical Subsystem Breakdown

- **Frontend:** Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS, Recharts, and Canvas visualizer.
- **Backend API:** FastAPI 0.109 with Uvicorn ASGI production server, Pydantic validation, and OAuth2 JWT auth.
- **AI / Deep Learning:** PyTorch 2.0+ and Ultralytics YOLOv8 Nano (`ml/models/best.pt`, 5.96 MB) for defect localization.
- **Database & DevOps:** PostgreSQL 15, SQLite 3, SQLAlchemy ORM, Docker Compose, and Nginx reverse proxy.

5. SYSTEM MODULES

5.1 Core Implemented Modules

The backend and machine learning subsystems are partitioned into seven cohesive, loosely coupled operational modules:



Figure 5.1: Interconnected System Modules Architecture

5.2 Module Responsibilities & Implementation

- **Auth & Ingestion** (``auth.py``, ``image_processing.py``): JWT token verification, RBAC permissions, and image quality gating.
- **AI/ML Subsystem** (``pipeline.py``): YOLOv8 defect localization, IoU NMS (0.65), and 73-class patch classification.
- **Severity & Quality Decisions** (``severity_engine.py``, ``assessment_engine.py``): Multi-factor severity math and 4-state routing.
- **API & Analytics** (``inspections.py``, ``analytics.py``, ``reports.py``): Run inspections, manual overrides, and KPI reporting.

6. DATASET AND DATA PREPROCESSING

6.1 Industrial MVTec Anomaly Detection (AD) Benchmark

VisionInspect AI is trained and validated on the industrial **MVTec AD benchmark**, covering 15 categories: 10 rigid objects (Bottle, Cable, Capsule, Hazelnut, Metal Nut, Pill, Screw, Toothbrush, Transistor, Zipper) and 5 textures (Carpet, Grid, Leather, Tile, Wood), mapped to 73 defect classes in ``class_mapping.json``.

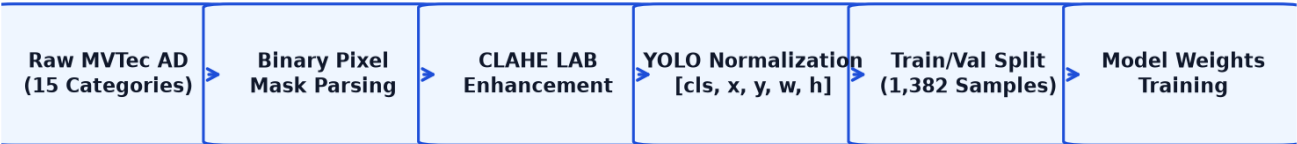


Figure 6.1: Dataset Preprocessing and Bounding Box Annotation Pipeline

6.2 Preprocessing & YOLO Dataset Transformation

- **Mask-to-BBox Conversion:** Pixel anomaly masks are converted into YOLO coordinates ``[cls, x_c, y_c, w, h]``.
- **Adaptive Contrast (CLAHE):** Enhances Luminance in LAB color space to reveal subtle surface scratches.
- **Data Augmentation:** Affine shifts, rotations ($\pm 15^\circ$), and noise injection ensure illumination and vibration invariance.

7. AI/ML MODEL AND INFERENCE PIPELINE

7.1 Deep Learning Detection & Classification

The computer vision subsystem utilizes an Ultralytics YOLOv8 Nano architecture (`ml/models/best.pt`, 5.96 MB) optimized for high-speed edge inference on normalized 640×640 tensors with a confidence threshold of 0.25.



Figure 7.1: Deep Learning Inference and Defect Disambiguation Pipeline

7.2 Model Inference & Deduplication Workflow

- **YOLO Forward Pass:** Computes feature maps to detect anomalous bounding box candidates and confidences.
- **IoU Deduplication:** Non-Maximum Suppression ($\text{IoU} \geq 0.65$) suppresses overlapping boxes on the same flaw.
- **Secondary Patch Classifier:** Evaluates localized bounding box crops to disambiguate fine-grained defect classes.
- **Deterministic Pipeline Hand-off:** Feeds bounding boxes, areas, flaw types, and confidences to the Severity Engine.

8. SEVERITY ASSESSMENT AND QUALITY DECISIONS

8.1 Multi-Factor Severity Scoring Equation

Total Severity Score = (Size Score × 0.30) + (Location Score × 0.25) + (Type Hazard Score × 0.25) + (Confidence Score × 0.20)

PASS Zero defects detected. Meets all quality criteria.	FAIL Critical uncorrectable flaw (Crack/Hole/Missing/Broken).	REVIEW Low confidence (<70%) or blurry image (POOR).	REWORK Correctable surface flaw (Scratch/Spot/Bent/Glue).
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Figure 8.1: Strict 4-State Manufacturing Quality Decision Routing

8.2 Deterministic Quality Decision Routing

- **PASS:** Zero defects detected. Product satisfies all optical and dimensional tolerance criteria.
- **FAIL:** Uncorrectable structural flaw (crack, hole, broken, missing) or composite severity ≥ 80.0.
- **REVIEW:** Detection confidence < 70%, classifier confidence < 60%, or image quality evaluated as POOR.
- **REWORK:** Correctable flaw (scratch, contamination, bent wire, glue residue) with acceptable confidence.

9. IMPLEMENTATION AND MILESTONES

9.1 Engineering Progression across Milestones 1–4

The project evolved systematically through four development milestones, progressing from foundational backend API and data schemas to computer vision integration, UI dashboards, and production containerization:

Milestone 1 Project Foundation • FastAPI REST Core • Relational DB Models • JWT Authentication	Milestone 2 AI Vision Engine • YOLOv8 Defect Model • Severity Scoring Math • Bounding Box Pipeline	Milestone 3 UI & Analytics • Next.js 16 Visualizer • 73-Class Taxonomy • Pareto & Trend Charts	Milestone 4 Testing & Deploy • Strict 4-State Decisions • 25/25 Tests Passed • Docker Containerization
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Figure 9.1: Chronological Milestone Roadmap & Deliverables

9.2 Milestone Deliverables & Verification

- **Milestone 1 (Foundation):** Engineered FastAPI backend, modeled SQLAlchemy tables, and built JWT auth.
- **Milestone 2 (AI Vision):** Integrated YOLOv8 model, built severity calculation math, and bounding box pipelines.
- **Milestone 3 (UI & Analytics):** Mapped 73 classes (`class_mapping.json`), built Next.js UI and Recharts dashboards.
- **Milestone 4 (Production):** Finalized 4-state routing, manual overrides, Docker stack, and verified 25/25 Pytests.

10. USER ROLES AND DASHBOARDS

10.1 Role-Based Access Control (RBAC)

VisionInspect AI enforces strict Role-Based Access Control across four authenticated user tiers:

ADMIN Full platform governance, user provisioning, catalog & model hot-swapping	QUALITY_ENG Runs inspection scans, reviews defect overlays, executes manual overrides	SUPERVISOR Monitors pass/fail rates, reviews 4-color quality trends, exports audit reports	OPERATOR Floor image ingestion, triggers camera capture, views instant PASS/FAIL
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Figure 10.1: Role-Based Access Control & Operational Permissions

10.2 Role Responsibilities & Dashboard Capabilities

- **ADMIN:** System governance, user provisioning, product catalog management, and model weight hot-swapping.
- **QUALITY_ENGINEER:** Real-time inspection viewer, defect bounding box inspection, and audited manual overrides.
- **SUPERVISOR:** Shift-level KPI monitoring, 4-color quality trend analysis, and CSV/PDF compliance report exports.
- **OPERATOR:** Floor inspection image upload and immediate visual status indicator (PASS / FAIL / REVIEW / REWORK).

11. END-TO-END INSPECTION WORKFLOW

11.1 High-Speed Execution Lifecycle

The inspection lifecycle executes a continuous 7-stage automated pipeline designed for sub-100ms response times:

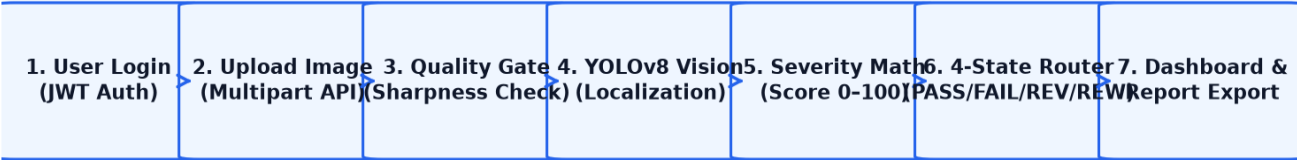


Figure 11.1: Complete Inspection Execution Lifecycle from Ingestion to Reporting

11.2 Detailed Lifecycle Operations

- **1. Ingestion & Quality Gate:** Uploads image via JWT-authenticated API; OpenCV verifies sharpness and exposure.
- **2. AI Vision & Classification:** YOLOv8 detects bounding boxes, IoU NMS deduplicates, and classifier maps 73 classes.
- **3. Severity & 4-State Routing:** Multi-factor severity math assigns PASS, FAIL, REVIEW, or REWORK routing.
- **4. Presentation & Reports:** Canvas visualizer renders overlays and persists inspection records in real time.

12. TESTING AND PERFORMANCE EVALUATION

12.1 Automated Testing Verification (25/25 Passed)

The codebase was verified via **25 automated Pytest unit and integration tests (100% pass rate)** in ``backend/tests``, validating API liveness, severity math ($\$S\times0.3+L\times0.25+...\$$), 4-state decisions, 73-class mappings, and Next.js builds.

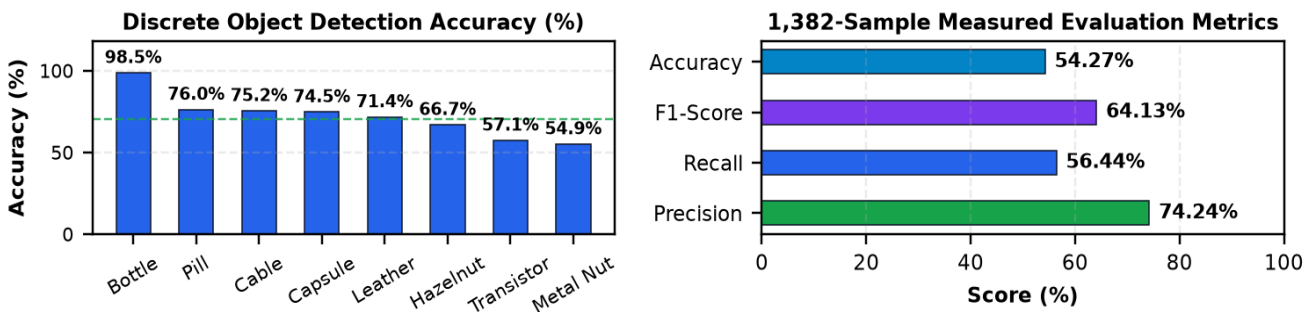


Figure 12.1: Empirical AI Evaluation on 1,382 MVTec AD Validation Samples

12.2 Measured Evaluation Results & Insights

- **Measured Metrics (1,382 Samples):** Precision = 74.24%, Recall = 56.44%, F1-Score = 64.13%, Accuracy = 54.27%.
- **High Performance on Discrete Objects:** Bottle (98.46%), Pill (76.00%), Cable (75.21%), Capsule (74.49%), Leather (71.43%).
- **Inference Latency:** YOLO tensor forward pass executes in <60ms on GPU and <100ms on standard CPU.

13. DEPLOYMENT AND CLOUD ARCHITECTURE

13.1 Multi-Container Docker Architecture

VisionInspect AI is fully containerized using Docker and Docker Compose, providing environment parity between local development, edge industrial PCs, and cloud VM hosts:

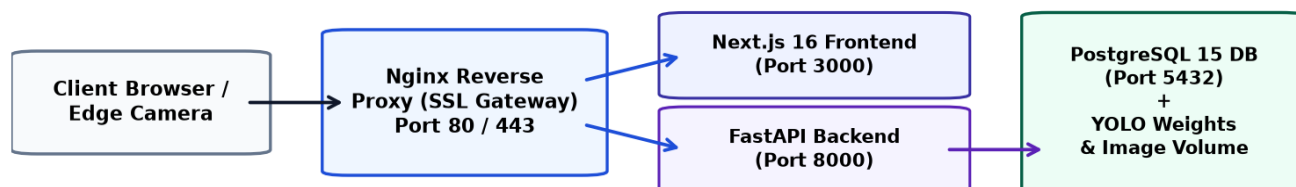


Figure 13.1: Production Container Network and Single-Origin Nginx Proxy Topology

13.2 Deployment Status & Cloud Readiness

- **Zero-CORS Reverse Proxy:** Nginx unifies Next.js (Port 3000) and FastAPI (Port 8000) under a single origin.
- **Local & Docker Deployment: Implemented & Verified.** Multi-container stack passes all health checks.
- **AWS Cloud Topology: Target Architecture Specified** for containerized hosting on AWS EC2 / Lightsail.
- **Microsoft Azure Assessment: Documented as Blocked** due to subscription limits on the designated account.

14. CONCLUSION AND FUTURE ENHANCEMENTS

14.1 Project Achievements & Industrial Value

VisionInspect AI successfully bridges deep computer vision and industrial quality operations. By uniting YOLOv8 defect localization with deterministic multi-factor severity scoring and strict 4-state routing, the system eliminates human inspection fatigue, prevents defect escapes, and minimizes scrap waste through automated REWORK routing.

Phase 1: Precision • YOLOv8 Polygon Seg • 1024x1024 High-Res • Texture optimization	Phase 2: Edge Acceleration • NVIDIA Jetson TensorRT • Sub-15ms inference • Direct GigE cameras	Phase 3: Industry 4.0 • PLC / Modbus bus • Robotic ejection arm • Explainable Grad-CAM
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Figure 14.1: Prioritized 3-Phase Future Roadmap & Industrial Extensions

14.2 Prioritized Future Roadmap

- **Phase 1 (Precision):** Integrate YOLOv8-Seg polygon segmentation for textures and adopt 1024×1024 tensors.
- **Phase 2 (Edge Acceleration):** Deploy TensorRT-compiled models on NVIDIA Jetson Orin edge modules (<15ms).
- **Phase 3 (Industry 4.0):** Implement PLC/SCADA Modbus bus integration for automated pneumatic part sorting.

THANK YOU

VisionInspect AI — Manufacturing Quality Intelligence System